

ECE148: Signal Processing Applications

Assignment 7: Range and Bearing-Angle Estimation

Andy Chen

May 2026

Part A: Bearing-Angle Estimation

A passive detector consists of two acoustic receivers. The separation between the receivers is *2.0 meters*. The acoustic propagation speed in air is *343.6 m/s*. The sampling rate of the A/D conversion is *48 kHz*.

Cross-Correlation and Time-Delay Estimation

We perform cross-correlation of the two data tracks. Cross-correlation measures the similarity between the two signals as one signal is shifted relative to the other. The discrete cross-correlation is defined as:

$$R_{xy}[k] = \sum_n x[n] y[n - k]$$

where $x[n]$ and $y[n]$ are sampled acoustic signals, and k is the lag variable in samples. The peak of the cross-correlation function corresponds to the lag value that produces the greatest similarity between the two signals. Because the lag is measured in samples, it must be converted into a physical time delay using the sampling frequency $f_s = 48 \text{ kHz}$:

$$\tau = \frac{k}{f_s}$$

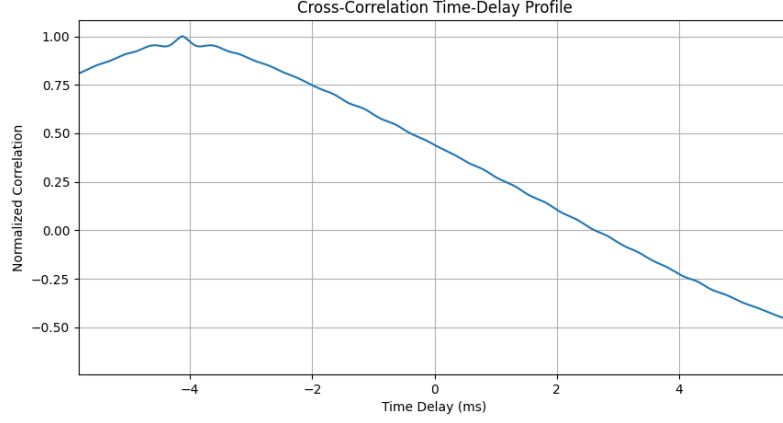


Figure 1: Plot of the resultant correlation function as the time-delay profile (estimate of the time-delay).

Bearing Angle Estimation

We convert the time delay estimation into a bearing angle using the geometry of the receiver arrangement. For a receiver spacing d and sound propagation speed c , the relationship between the time delay τ and bearing angle θ is:

$$\tau = \frac{d \sin(\theta)}{c}$$

Solving for the bearing angle provides:

$$\theta = \sin^{-1} \left(\frac{c\tau}{d} \right)$$

Only delays satisfying $|\frac{c\tau}{d}| = 1$ are physically valid, since the argument of inverse sine must remain within the interval $[-1, 1]$.

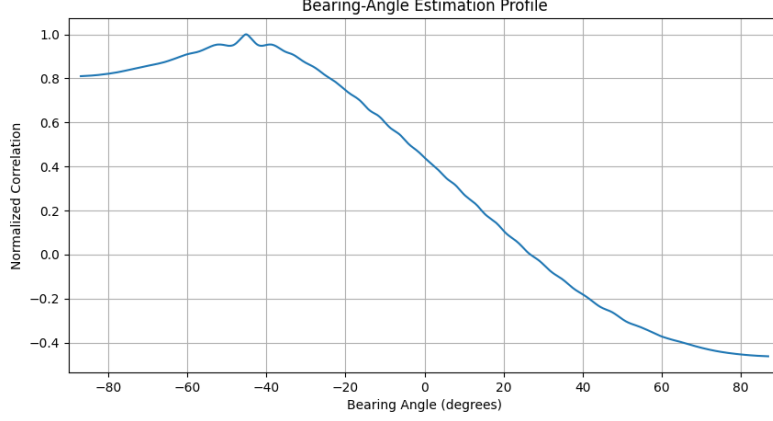


Figure 2: Plot of the bearing-angle estimation profile of the acoustic source as a function of the angle.

Part B: 5-Gun Salute

Given a single-channel (mono) soundtrack, we produce a stereo soundtrack such that, with headphones, the sound source moves from -60° to $+60^\circ$ in 5 steps, at -60° , -30° , 0° , $+30^\circ$, and $+60^\circ$ sequentially.

Approach and Implementation

The implementation uses interaural time difference (ITD) to simulate binaural localization cues. Essentially, human listeners localize sound partly by detecting small differences in arrival time between the two ears. If a source is located on the left side, the sound reaches the left ear slightly earlier, and reaches the right ear slightly later. Similarly, right-side sources arrive earlier at the right ear.

For a source located at angle θ , the interaural delay is approximately:

$$\tau = \frac{d \sin(\theta)}{c}$$

where d is the ear spacing, c is the speed of sound, and θ is the source angle. We implement this delay digitally by delaying one stereo channel relative to the other.

Implementation and Results

Assuming an ear spacing of $d = 0.20$ m, we compute the interaural delay τ for each angle θ . The delay was then converted into samples $N = \tau f_s$ where f_s is the sampling frequency. For negative angles, the right channel was delayed.

For positive angles, the left channel was delayed. At zero degrees, no delay was applied. The stereo version of the entire audio sample was generated independently for each angle. The five processed stereo signals were then concatenated sequentially to form the final stereo output.

The generated soundtrack successfully created the perception of a moving sound source when played through headphones. Each repetition of the audio sample was played at the distinct locations $\theta \in [-60^\circ, -30^\circ, 0, +30^\circ, +60^\circ]$. The localization effect was produced entirely through interaural time difference cues. The delayed arrival at one ear caused the listener’s auditory system to interpret the sound as originating from the corresponding direction.

Part C: Range Estimation

We perform step-frequency FMCW radar imaging using FFT for range estimation. The given dataset was taken over a section of the walkway pavement in front of Broida Hall. The ground-penetrating radar imaging unit scanned along a linear path and took data at 200 spatial positions. The spatial spacing between the data-collection points was 2.13 cm. At each data-collection position, the system illuminated the subsurface region of the walkway area with microwaves in the step-frequency (FMCW) mode, stepping through 128 frequencies with a constant increment, from 0.976 GHz to 2.00 GHz. The relative permittivity ϵ_r is approximately 6.0. The dataset is in the form of a 200×128 array, corresponding to 128 complex-amplitude values (for the 128 frequencies) at 200 data-acquisition points.

Signal Processing Procedure

The original radar data has dimensions:

$$(200 \times 128)$$

At each of the 200 spatial positions, the 128 frequency samples were transformed into depth profiles using an inverse Fourier transform. To improve depth sampling density, zero-padding was first applied along the frequency dimension, resulting in a zero-padded data set of size:

$$(200 \times 1024)$$

A total of 200 inverse FFT operations of length 1024 were then performed, one for each spatial acquisition position. This process converts the stepped-frequency radar measurements into range profiles. After the range transformation, only the shallow-depth region was retained because the rebars are located near the surface. Therefore, the range profiles were truncated from 1024 samples down to the first 43 depth bins, producing a new data array of size:

$$(200 \times 43)$$

The physical depth axis was calculated using:

$$d = \frac{vt}{2}$$

where $v = \frac{c}{\sqrt{\epsilon}} = 1.225 \times 10^8$ m/s is the propagation speed in the pavement, and the factor of two accounts for the round-trip propagation. The frequency bandwidth is:

$$B = 2.00 \text{ GHz} - 0.96 \text{ GHz} = 1.024 \text{ GHz}$$

Given the truncated range-profile data with dimensions (200×43) , we improve smoothness and sampling density in the depth direction by performing FFT-based zero-padding. For each 43-point range profile, a Fourier transform was computed, zeros were inserted into the spectral domain to expand the sequence from 43 samples to 128 samples, and a 128-point inverse FFT was performed. This resulted in an interpolated data set of size:

$$(200 \times 128)$$

We then apply FFT-based interpolation along the horizontal direction. The aperture dimension was first zero-padded:

$$(256 \times 128)$$

In the spatial dimension, we applied FFT interpolation using zero-padding and inverse FFT processing, producing a final image size of:

$$(2048 \times 128)$$

Because radar reflections span a large dynamic range, we applied logarithmic compression to the image. This improved the visibility of weaker reflections, such as the rebars. The display range was clipped from -40 to 0 dB to improve image contrast.

Resolution Limit and Maximum Detectable Depth

The depth resolution of the FMCW radar system is determined by the total bandwidth:

$$\Delta d = \frac{v}{2B}$$

Substituting the propagation velocity and bandwidth, we have:

$$\Delta d \approx 0.0598 \text{ m}$$

Thus, the depth resolution limit is approximately 6.0 cm.

We calculate the pixel spacing using interpolation ratios starting with the depth resolution limit. After extending the data set to (200×1024) , we find the pixel size to be:

$$\Delta y = \frac{5.98 \text{ cm}}{8} = 0.748 \text{ cm}$$

After further interpolation, we find the new pixel size to be approximately:

$$\Delta y = \frac{0.748 \text{ cm}}{3} = 0.249 \text{ cm}$$

The maximum depth is therefore:

$$0.249 \times 128 = 31.895 \text{ cm}$$

Physical Pixel Size

The original horizontal spacing is:

$$\Delta x = 2.13 \text{ cm}$$

The interpolation ratio in the horizontal direction is:

$$\frac{1024}{128} = 8$$

We find that the pixel size in the horizontal direction is:

$$\Delta x = \frac{2.13 \text{ cm}}{8} = 0.266 \text{ cm}$$

In the vertical direction, we found the depth resolution limit to be approximately 6.0 cm. Dividing by the interpolation ratio, we find:

$$\Delta y = \frac{5.98 \text{ cm}}{8} = 0.748 \text{ cm}$$

However, after truncating the array to (1024×43) , we find the interpolation ratio to be:

$$\frac{128}{43} \approx 3$$

Therefore, the final pixel spacing in the depth direction is:

$$\Delta y = \frac{0.748}{3} = 0.249 \text{ cm}$$

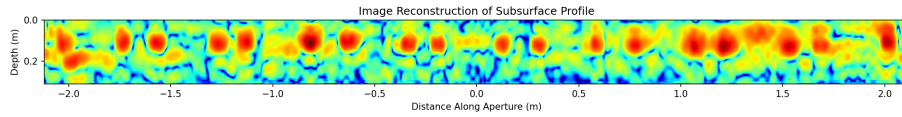


Figure 3: Image reconstruction of the subsurface profile.

Summary

We performed step-frequency FMCW radar imaging using FFT-based range reconstruction. The frequency-domain radar measurements were converted into depth profiles using inverse FFT processing, with propagation speed corrected using the relative permittivity of the pavement material. FFT-based interpolation was applied in both directions to produce a visually smooth image with approximately square pixels. The resulting radar image successfully reconstructs shallow subsurface features within the walkway structure.